

MCGREGOR, MN, USA

WMO: 720258

Lat: 46.619N

Lon: 93.310W

Elev: 1228

StdP: 14.06

Time Zone: -6.00 (NAC)

Period: 04-19

WBAN: 04997

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -17.4	(c) -12.9	(d) -24.1	(e) 1.5	(f) -15.8	(g) -18.3	(h) 2.1	(i) -10.5	(j) 20.1	(k) 14.5	(l) 18.5	(m) 15.7	(n) 3.3	(o) 310	(p) 0.585

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.3	85.8	71.4	81.9	68.1	80.6	67.1	73.6	82.6	71.5	79.3	69.3	75.7	9.8	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
70.5	117.4	78.9	68.4	109.2	76.4	66.4	101.7	74.4	38.0	82.1	36.0	80.6	34.1	76.9	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 18.9	(b) 17.0	(c) 15.6	(e) -25.5	(f) 90.2	(g) 5.6	(h) 2.8	(i) -29.5	(j) 92.2	(k) -32.8	(l) 93.8	(m) -35.9	(n) 95.4	(o) -40.0	(p) 97.4		
(a) 18.9	(b) 17.0	(c) 15.6	(e) -25.8	(f) 77.4	(g) 5.5	(h) 2.4	(i) -29.7	(j) 79.1	(k) -33.0	(l) 80.4	(m) -36.0	(n) 81.8	(o) -40.1	(p) 83.5		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
Temperatures, Degree-Days and Degree-Hours	DBAvg	41.6	12.3	15.0	29.0	41.5	53.0	62.4	67.9	64.7	57.7	44.8	31.4	17.5	
	DBStd	21.52	13.76	12.05	12.23	9.11	7.99	6.11	5.65	5.69	7.92	8.08	10.07	11.82	
	HDD50	4938	1169	981	657	279	57	2	0	0	20	206	560	1007	
	HDD65	8814	1634	1401	1116	705	382	118	34	76	242	627	1007	1472	
	CDD50	1861	0	0	6	23	150	373	555	455	252	44	3	0	
	CDD65	262	0	0	0	0	10	39	124	65	23	1	0	0	
	CDH74	2719	0	0	0	3	184	461	1225	620	210	16	0	0	
	CDH80	536	0	0	0	0	38	69	288	112	28	1	0	0	
Wind	WSAvg	6.5	6.4	6.5	6.7	7.8	7.2	6.3	5.4	5.1	6.2	6.7	6.9	6.3	
Precipitation	PrecAvg	28.8	0.7	0.7	1.2	2.2	3.1	4.4	4.6	3.4	3.2	3.2	1.3	0.8	
	PrecMax	35.2	2.0	1.6	2.9	6.9	5.7	7.2	7.9	7.7	6.3	7.9	3.3	1.5	
	PrecMin	19.3	0.2	0.1	0.3	0.7	1.5	1.3	2.4	1.3	0.9	0.6	0.2	0.2	
	PrecStd	4.1	0.5	0.4	0.7	1.3	1.2	1.5	1.8	1.7	1.4	1.8	0.9	0.4	
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	42.5	45.9	67.9	72.5	85.7	86.0	90.1	87.8	83.6	76.8	63.0	42.9	
		MCWB	36.7	40.1	59.6	55.7	67.0	71.4	74.7	73.4	69.7	60.3	52.6	38.8	
	2%	DB	36.2	39.1	57.4	67.9	79.2	81.5	86.3	82.2	79.5	68.5	54.7	37.1	
		MCWB	33.9	35.3	48.8	52.5	61.7	67.0	72.2	69.3	66.9	57.7	47.5	34.6	
	5%	DB	33.7	35.7	50.2	63.2	73.2	79.2	82.2	80.7	76.7	63.2	50.1	34.4	
		MCWB	32.8	33.0	43.8	49.4	58.2	64.6	69.1	67.6	65.6	53.8	44.0	33.3	
	10%	DB	31.7	32.0	44.8	58.6	69.8	76.9	80.9	77.3	72.2	58.7	45.4	32.4	
		MCWB	30.9	30.3	39.5	46.7	56.8	63.1	67.9	65.4	63.4	51.1	41.5	31.8	
	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	37.2	41.3	59.6	58.5	71.1	73.5	77.3	74.6	72.8	63.7	54.4	40.3
			MCDB	42.6	46.8	65.2	68.3	79.6	82.4	85.3	83.5	80.4	70.2	59.4	41.7
2%		WB	34.2	35.8	49.8	54.0	64.9	70.2	74.5	72.3	69.8	59.2	48.4	35.5	
		MCDB	35.8	38.4	56.9	63.6	74.0	78.0	83.6	80.4	75.3	67.4	54.1	37.3	
5%		WB	32.2	32.6	43.9	50.3	61.8	67.5	72.2	70.2	67.4	55.2	44.8	33.5	
		MCDB	33.0	34.4	49.7	61.3	70.4	76.2	80.6	76.4	74.2	61.6	48.3	34.2	
10%		WB	30.2	30.2	39.7	47.3	58.5	65.2	69.8	68.0	64.6	52.0	41.5	31.9	
		MCDB	31.0	31.6	44.3	57.6	67.3	72.9	76.6	74.9	70.3	56.7	46.1	32.4	
Mean Daily Temperature Range	5% DB	MDBR	17.0	18.7	18.4	19.8	22.2	21.6	21.3	21.4	20.6	16.5	13.8	13.8	
		MCDBR	15.5	18.3	23.2	29.4	30.3	25.9	23.5	24.3	25.2	25.2	21.4	13.6	
		MCWBR	13.3	14.6	14.9	16.7	15.5	13.4	11.6	12.6	14.3	14.7	14.0	12.0	
	5% WB	MCDBR	14.7	16.9	21.9	26.7	24.8	22.2	21.4	20.2	21.0	21.2	19.3	12.4	
		MCWBR	13.2	13.8	14.7	16.3	14.8	13.4	12.1	12.0	13.3	14.2	13.5	11.4	
Clear-Sky Solar Irradiance	taub	0.248	0.262	0.305	0.359	0.380	0.378	0.379	0.386	0.358	0.309	0.285	0.260		
	taud	2.391	2.359	2.374	2.340	2.334	2.365	2.381	2.382	2.436	2.569	2.515	2.415		
	Ebn at Noon	275	293	293	285	281	281	279	272	272	273	259	256		
	Edh at Noon	22	28	32	37	39	38	36	35	30	23	20	19		
All-Sky Solar Radiation	RadAvg	495	850	1205	1475	1689	1834	1955	1636	1234	754	465	351		
	RadStd	54	74	116	144	127	125	79	125	92	100	49	33		

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
Station Only		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (52 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page